Xylem Pipeline Relay

Move water uphill like a tree, faster and cleaner than rival teams.

Win condition: Most water delivered with speed, low spill, and strong explanation.

THE SCIENCE YOU ARE USING

Capillary action. Water climbs narrow spaces on its own. Water molecules stick to a surface (adhesion) and to each other (cohesion), so a thin column is pulled upward against gravity. Narrower channels and more wettable materials pull water higher and faster. Trees use this in their xylem.

Material and geometry. A felt strip, a cotton wick, and a folded paper towel each move water at a different rate. Path length, slope, and contact area all matter. Your team picks the material and the route to deliver the most water to a target cup.

HOW TO BUILD AND RUN IT

01 **Test three wick materials.** Dip paper towel, cotton, and felt into colored water. Watch which climbs fastest and highest.

02 **Pick a material and route.** Choose your wick and lay out a path from the source cup up over a block to the target cup.

03 **Run the first transfer.** Start the wick and time how much water reaches the target in the time limit. Catch spills in the tray.

04 **Reduce spill and length.** Trim the path, improve contact, and re-run. Less spill and a shorter path usually deliver more.

05 **Final delivery round.** Run the best design for score. Water delivered, speed, and low spill all count.

Safety: Use trays. Wipe spills immediately. No standing on chairs or stacked furniture. Allergy: Food coloring may stain. Use washable dye and gloves if needed.

MATERIALS

Paper towels	shared
Cotton string	shared
Felt strips	shared
Cups	per team
Food coloring	shared
Trays	per team
Elevation blocks	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ____________________

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The ONE variable we are testing: ____________________

Baseline result: ____________________

After redesign: ____________________

DEFEND YOUR DESIGN

What evidence made you change your design? ____________________

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If you ran it again, what is the ONE thing you would change? ____________________

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HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Water delivered	40
Speed	20
Least spill	15
Design explanation	15
Redesign gain	10
Total	**100**